

5.2.2 Enthalpy and entropy

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
Entropy	
(a) explanation that entropy is a measure of the dispersal of energy in a system which is greater, the more disordered a system	HSW1 The model of entropy to explain thermodynamic stability.
(b) explanation of the difference in magnitude of the entropy of a system: (i) of solids, liquids and gases (ii) for a reaction in which there is a change in the number of gaseous molecules	
(c) calculation of the entropy change of a system, ΔS , and related quantities for a reaction given the entropies of the reactants and products	M2.2, M2.3, M2.4
Free energy	
(d) explanation that the feasibility of a process depends upon the entropy change and temperature in the system, $T\Delta S$, and the enthalpy change of the system, ΔH	HSW1,5,6 Use of entropy, enthalpy and temperature for predicting feasibility.
(e) explanation, and related calculations, of the free energy change, ΔG , as: $\Delta G = \Delta H - T\Delta S$ (the Gibbs' equation) and that a process is feasible when ΔG has a negative value	M0.0, M2.2, M2.3, M2.4 HSW5 Link between ΔG and feasibility.
(f) the limitations of predictions made by ΔG about feasibility, in terms of kinetics.	M0.3 HSW6 The relative effects of entropy and rate in determining feasibility of processes.